

Atypical handedness as a marker of altered brain lateralization: genetic and clinical study in bipolar disorder

Inés García-Ortiz¹, Lucía Albarracín García^{2,3,4,5,6,7}, Elisabet Madrdejos Palomares^{3,4}, Miriam Martínez-Jiménez¹, Rosa Ana Bello Sousa⁴, Inés Carpio-López¹, Sergio Sánchez-Alonso⁴, Sergio Benavente López⁵, Raquel Álvarez García⁶, Laura Jiménez Muñoz^{3,4,5}, Ezequiel Di Stasio⁷, Carlos López de la Hoz¹, Raúl Somavilla Cabrero¹, Noelia Borge García¹, Rebeca Jiménez⁸, Marta Bermejo Bermejo⁸, Janice Fullerton^{9,10}, Enrique Baca García^{3,4,5,6,7,11,12,13}, Silvia Paracchini¹⁴, Claudio Toma^{1,15}

¹ Centro de Biología Molecular Severo Ochoa (CBMSO), UAM/CSIC, Madrid, Spain

² School of Psychology, Universidad Alfonso X el Sabio, Madrid, Spain

³ Instituto de Investigación Sanitaria Fundación Jiménez Díaz, Madrid, Spain

⁴ Department of Psychiatry, Hospital Universitario Fundación Jiménez Díaz, Madrid, Spain.

⁵ Department of Psychiatry, Hospital Universitario Infanta Elena, Valdemoro, Madrid, Spain.

⁶ Department of Psychiatry, Hospital Universitario Rey Juan Carlos, Mostoles, Madrid, Spain.

⁷ Department of Psychiatry, Hospital General de Villalba, Madrid, Spain.

⁸ Department of Occupational Medicine and Work Health, Consejo Superior de Investigaciones Científicas (CSIC), Madrid, Spain.

⁹ Neuroscience Research Australia, Sydney, NSW, Australia

¹⁰ School of Biomedical Sciences, University of New South Wales, Sydney, NSW, Australia

¹¹ Department of Psychiatry, Universidad Autónoma de Madrid, Madrid, Spain.

¹² CIBERSAM (Centro de Investigación Biomédica en Red Salud Mental), Instituto de Salud Carlos III, Madrid, Spain

¹³ Universidad Católica del Maule, Talca, Chile

¹⁴ School of Medicine, University of St Andrews, St Andrews, UK.

¹⁵ School of Clinical Medicine, Discipline of Psychiatry and Mental Health, University of New South Wales, Sydney, NSW, Australia

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Background: Handedness is associated with brain asymmetries. The prevalence of non-right-handedness (left-handedness, mixed-handedness, and ambidexterity) is estimated at 18% in the general population but has been reported at higher rates in the psychiatric conditions like schizophrenia (SCZ), suggesting altered patterns of brain asymmetries in these conditions. Studies of non-right-handedness in bipolar disorder (BD) are limited, and frequently rely on small cohorts or lack of deep phenotypic data.

Objectives: We aim to: i) estimate the prevalence of non-right-handedness in BD, and across subtypes and clinical features; ii) evaluate polygenic score (PGS) for handedness, schizophrenia and BD across diagnostic categories, and handedness groups; iii) identify differential gene expression associated with handedness using blood-derived imputed transcriptomic data across 13 brain tissues.

Methods: The cohort comprises 380 BD cases and 1,600 controls with Edinburgh Handedness Inventory (EHI) data from the MadManic cohort. Clinical data includes psychosis, BD subtypes, suicide behaviour, and severity index measured by number of hospitalizations, mania and depressive events. PGS will be computed with PRScsx based on large-scale GWAS of handedness (N = 1,766,671), BD (N = 840,309) and SCZ (N = 161,405). RNA sequencing (RNAseq) data are available for 720 subjects with EHI data. Brain tissue-specific gene expression will be imputed using BrainGenie and analyzed with limma.

Results: We anticipate completing most of those analyses and present the results of handedness in BD compared to controls, stratified by symptoms and severity. The polygenic and gene expression analyses have the potential to shed light on the underlying mechanisms potentially differentiating between shared and specific effects compared to other conditions such as SCZ.

Conclusion: This study, which will be one of the largest of this kind conducted to date for BD, will provide a comprehensive characterization of handedness in BD by integrating phenotypic, genetic, and transcriptomic data. Findings may clarify the role of altered brain lateralization as a transdiagnostic marker across severe mental disorders.